
Sheet 2

Question 1:

A waveguide is formed by a material of index $n=3.5$, a step index $\Delta n=0.1$ and the guide thickness is $2d = 1$ microns. Calculate, at the wavelength of 1550 nm:

- 1- The effective index of the fundamental guided mode
- 2- The number of modes in the guide
- 3- The guide numerical aperture NA.

Question 2:

A step index single mode waveguide is designed to have a V number of 2 at the wavelength of 1300 nm with a numerical aperture of 0.12 and a core index of 1.5. If the guide is now operating at the wavelength of 650 nm, calculate:

- a- The number of modes in the guide at 650 nm wavelength
- b- The waveguide width d
- c- The propagation constants of the highest order mode and the lowest order mode in the guide at the wavelength $\lambda = 650$ nm.

Question 3:

A symmetric slab waveguide has the following parameters: $d=560$ nm, $n_1 = 1.5$ and $n_2=1.48$. What is the effective waveguide thickness for the fundamental mode at the wavelength $\lambda=630$ nm and $\lambda = 850$ nm?

Question 4:

A dielectric optical waveguide is formed by an index change of 0.01 in a material of substrate index 1.5. Calculate the guide thickness such that its normalized frequency is $V = 3$ at the wavelength of 1300 nm. If the guide is excited by the fundamental mode of another waveguide of the same index change and a thickness of $4.5 \mu\text{m}$ and the axis of the two waveguides are aligned together with the amplitude of the electric field of this mode is 1V/mm, calculate the amplitude of the electric field of the fundamental mode of the 1st guide then calculate the power in this mode.

Question 5:

A surface slab waveguide is fabricated on a glass substrate of refractive index $n=1.5$ by using the technology of the ionic exchange. If the resultant index change is assumed to be a step index of $\Delta n=5 \times 10^{-3}$, calculate the maximum guide thickness to keep a single mode operation in the guide at the wavelength $\lambda=1300$ nm.

Question 6:

Consider a symmetric planar guide with $n_1=1.5$, $n_2=1.48$, and a thickness $d=3.912 \mu\text{m}$. Calculate: a- The number of guided modes at $\lambda=1 \mu\text{m}$.

b- The propagation constants and effective refractive indices of these modes.

c- Assume that at the guide input $z=0$, the input electric field distribution is given by:

$$E_y(x) = 1.375 \times 10^4 \cos(h_0 x) + 1.309 \times 10^4 \sin(h_1 x)$$

with $h_0=0.5978 \mu\text{m}^{-1}$ and $h_1=1.1649821 \mu\text{m}^{-1}$, find:

- i- The power carried by each mode in the guide.
- ii- The electric field distribution at $z=0$, $58.9 \mu\text{m}$, and $117.88 \mu\text{m}$.

